Prove tangential derivative convergence of NSDP central path as barrier vanishes

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Website #20 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

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Problem

We are given a smooth primal–dual central path $\mu \mapsto w(\mu) = (x(\mu), Y(\mu), z(\mu))$ for the NSDP

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad G(x) \in \mathbb{S}^m_+,\ h(x) = 0,$$

satisfying, for all sufficiently small $\mu > 0$,

$$\nabla_x L(x(\mu), Y(\mu), z(\mu)) = 0, \qquad h(x(\mu)) = 0, \qquad G(x(\mu))Y(\mu) = \mu I,$$

with $G(x(\mu)) \in \mathbb{S}^m_{++}$, $Y(\mu) \in \mathbb{S}^m_{++}$, and $w(\mu) \to w_a = (x^*, Y_a, z_a)$ as $\mu \downarrow 0$.

The prompt assumes the paper's existence theory: under strict complementarity, enhanced SOSC (uniformly over the multiplier set), and MFCQ, there is a unique smooth central path converging to w_a, and

$$\frac{x(\mu) - x^*}{\mu} \to \xi^* \qquad (\mu \downarrow 0),$$

where ξ^* is the unique Δx-component of the paper's limit linearized symmetric BKKT system.

The question is whether one must have

$$\lim_{\mu \downarrow 0} \dot{x}(\mu) = \xi^*, \qquad \dot{x}(\mu) := \frac{d}{d\mu} x(\mu).$$

Alternative readings considered

The statement admits two natural readings.

- **Strong reading (adopted here).** The limit should hold under the exact weaker assumptions stated in the prompt, i.e., *without* the classical nondegeneracy condition.
- **Weaker reading.** The limit should hold under the older, stronger Yamashita–Yabe regime that includes nondegeneracy.

I work with the strong reading first. I do *not* fully settle it from the information in the prompt. What I do prove rigorously is:

1. the limit is affirmative in the classical nondegenerate regime;
2. the limit is also affirmative in an explicit *degenerate* family with nonunique multipliers, so the absence of nondegeneracy does not by itself destroy the conclusion.

Related literature found

- **Okuno (2025) — PARTIAL**. Exact setting of the prompt: the accessible abstract states existence and uniqueness of the central path under strong/enhanced SOSC, strict complementarity, and MFCQ, convergence of the dual multipliers to the analytic center of the multiplier set, and solvability of Newton equations near the path; the abstract does not state the $\dot{x}(\mu) \to \xi^*$ conclusion.
- **Yamashita–Yabe (2012) — PARTIAL/TOOL**. Stronger regime with nondegeneracy. I audited their accessible PDF at the point where they prove nonsingularity of the symmetrized Jacobian and then obtain a C^1 solution branch $\bar{w}(\mu)$ with $\bar{w}(0) = w^*$ by the implicit function theorem. This is the key tool for the strengthened result proved below.
- **Halická (2002) — BACKGROUND**. For linear SDP, strict complementarity alone yields analyticity of the central path at $\mu = 0$, hence all derivatives have finite boundary limits. This strongly suggests that boundary differentiability is plausible in semidefinite settings, but it does not directly answer the nonlinear problem here.
- **Jittorntrum (1984) — BACKGROUND**. A promising older reference on solution-point differentiability in nonlinear programming without strict complementarity. I did not use it in the proof because I could not audit the full theorem statement and its exact hypothesis match from the accessible web material alone.

Proof sketch

There are two distinct issues.

First, the implication

$$\frac{x(\mu) - x^*}{\mu} \to \xi^* \quad \Longrightarrow \quad \dot{x}(\mu) \to \xi^*$$

is *not* a purely calculus-level tautology. A scalar function such as $\mu^2 \sin(1/\mu)$ has vanishing secant quotient divided by μ, but its derivative oscillates. So one needs structural information from the BKKT system.

Second, under the *stronger* classical nondegeneracy hypothesis, the structure is enough: Yamashita–Yabe prove that the BKKT branch extends as a C^1 map to $\mu = 0$. Once this extension exists, the desired limit is immediate: the boundary derivative equals the limit of the secant quotients, and the prompt already identifies that secant limit with ξ^*.

The genuinely interesting part is the weaker, degenerate regime. I do not extract a complete proof from the prompt alone. However, I give a nontrivial explicit degenerate NSDP, with a non-singleton multiplier set and analytic-center limit multiplier, for which the central path can be solved exactly and the conclusion $\dot{x}(\mu) \to \xi^*$ holds identically. This shows that degeneracy is not an obstruction by itself.

Main result

Theorem 1. *The following two affirmative results hold.*

(A) ***Classical nondegenerate regime.*** *Assume, in addition to the prompt's hypotheses, the stronger Yamashita–Yabe assumptions: local Lipschitz continuity of second derivatives, the* *second-order sufficient condition, strict complementarity, and the nondegeneracy condition. Then*

$$\lim_{\mu\downarrow 0} \dot{x}(\mu) = \xi^*.$$

(B) ***Explicit degenerate family.*** *Fix integers $n \geq 2$ and $m \geq 2$. Write $x = (t, u) \in \mathbb{R} \times \mathbb{R}^{n-1}$, and consider*

$$f(t,u) := m\,t + \frac{1}{2}\|u\|^2, \qquad G(t,u) := tI_m, \qquad h \equiv 0.$$

Then $x^* := (0,0)$ is a KKT point whose multiplier set is non-singleton, the analytic center multiplier is $Y_a = I_m$, the problem satisfies MFCQ, strict complementarity, and the enhanced SOSC at x^*, and the unique central path is

$$x(\mu) = (\mu, 0), \qquad Y(\mu) = I_m.$$

Hence

$$\dot{x}(\mu) \equiv (1, 0, \ldots, 0) = \xi^*,$$

so in particular $\lim_{\mu\downarrow 0} \dot{x}(\mu) = \xi^*$.

Proof

Lemma 1. *Under the stronger Yamashita–Yabe hypotheses, the central path extends as a C^1 map to $\mu = 0$.*

Proof. Yamashita–Yabe prove the following: under assumptions (A1)–(A4) (their local Lipschitz second derivative assumption, SOSC, strict complementarity, and nondegeneracy), there exists a continuously differentiable function

$$\bar{w}(\mu) = (\bar{x}(\mu), \bar{y}(\mu), \bar{Z}(\mu))$$

defined for $\mu \geq 0$ sufficiently small such that

$$\bar{w}(0) = w^*, \qquad r(\bar{w}(\mu), \mu) = r_S(\bar{w}(\mu), \mu) = 0$$

for all such μ, with $\bar{X}(\mu) \succ 0$ and $\bar{Z}(\mu) \succ 0$ for $\mu > 0$.

Hypothesis audit. Their notation is the same BKKT system as in the prompt after renaming $g \mapsto h$ and $X \mapsto G$. In part (A) of the theorem I am explicitly assuming their (A1)–(A4), so every hypothesis matches verbatim.

Because the prompt assumes uniqueness of the central path for $\mu > 0$, the given $w(\mu)$ must coincide with $\bar{w}(\mu)$ for all sufficiently small $\mu > 0$. Therefore $x(\mu)$ extends to a C^1 map on $[0, \bar{\mu})$, with $x(0) = x^*$.

Hence

$$\lim_{\mu\downarrow 0} \dot{x}(\mu) = x'_+(0) = \lim_{\mu\downarrow 0} \frac{x(\mu) - x^*}{\mu}.$$

By the prompt's hypothesis, the right-hand secant quotient converges to ξ^*. Therefore

$$\lim_{\mu\downarrow 0} \dot{x}(\mu) = \xi^*.$$

$\square$

This proves part (A) of the theorem.

We now turn to the explicit degenerate family in part (B).

Lemma 2. *For the family*

$$f(t,u) = m\,t + \frac{1}{2}\|u\|^2, \qquad G(t,u) = tI_m, \qquad h \equiv 0,$$

the point $x^* = (0,0)$ is a KKT point with multiplier set

$$\Lambda(x^*) = \{Y \in \mathbb{S}^m_+ : \mathrm{tr}(Y) = m\}.$$

Its analytic center is $Y_a = I_m$.

Proof. At $x^* = (0,0)$ we have $G(x^*) = 0$, so primal feasibility holds. The stationarity condition reads

$$\nabla f(x^*) - DG(x^*)^*[Y] = 0.$$

Now $\nabla f(x^*) = (m,0)$, and since $DG(x^*)[(\tau,v)] = \tau I_m$, its adjoint is

$$DG(x^*)^*[Y] = (\mathrm{tr}\,Y, 0).$$

Thus stationarity is equivalent to $\mathrm{tr}\,Y = m$. Together with dual feasibility $Y \succeq 0$, this yields

$$\Lambda(x^*) = \{Y \succeq 0 : \mathrm{tr}\,Y = m\}.$$

To find the analytic center, maximize $\log\det Y$ over $Y \succ 0$ with $\mathrm{tr}\,Y = m$. If $\lambda_1, \ldots, \lambda_m > 0$ are the eigenvalues of Y, then $\sum_i \lambda_i = m$, and AM–GM gives

$$(\det Y)^{1/m} = (\lambda_1 \cdots \lambda_m)^{1/m} \le \frac{\lambda_1 + \cdots + \lambda_m}{m} = 1,$$

with equality if and only if $\lambda_1 = \cdots = \lambda_m = 1$. Hence the unique maximizer is

$$Y_a = I_m.$$

□

Lemma 3. *The same family satisfies MFCQ, strict complementarity, and the enhanced SOSC at $x^* = (0,0)$.*

Proof. **MFCQ.** Since $h \equiv 0$, only the semidefinite constraint matters. For the direction $d = (1,0)$,

$$DG(x^*)[d] = I_m \succ 0.$$

Hence the semidefinite MFCQ holds.

Strict complementarity. We have

$$G(x^*) = 0, \qquad Y_a = I_m \succ 0,$$

so $G(x^*) + Y_a = I_m \succ 0$. Thus strict complementarity holds.

Enhanced SOSC. The critical directions are (τ, v) such that

$$DG(x^*)[(\tau,v)] = \tau I_m \in T_{\mathbb{S}^m_+}(0) = \mathbb{S}^m_+,$$

and

$$\nabla f(x^*)^\top (\tau, v) = m\tau = 0.$$

Therefore $\tau = 0$, and the critical cone is

$$C(x^*) = \{(0, v) : v \in \mathbb{R}^{n-1}\}.$$

Moreover,

$$\nabla^2_{xx} L(x^*, Y, z) = \nabla^2 f(x^*) = \begin{pmatrix} 0 & 0 \\ 0 & I_{n-1} \end{pmatrix}$$

for every multiplier $Y \in \Lambda(x^*)$, because G is linear and $h \equiv 0$. Thus for any nonzero critical direction $(0, v)$,

$$(0, v)^\top \nabla^2_{xx} L(x^*, Y, z)\,(0, v) = \|v\|^2 > 0.$$

Since $DG(x^*)[(0, v)] = 0$, the extra curvature contribution appearing in the enhanced SOSC vanishes on the critical cone, so the enhanced SOSC holds uniformly over $\Lambda(x^*)$. □

Lemma 4. *For the same family, the unique central path is*

$$x(\mu) = (\mu, 0), \qquad Y(\mu) = I_m.$$

Proof. The BKKT conditions are

$$\nabla_x L((t, u), Y) = 0, \qquad tY = \mu I_m, \qquad t > 0,\ Y \succ 0.$$

Since

$$\nabla_x L((t, u), Y) = \big(m - \operatorname{tr} Y,\ u\big),$$

stationarity yields

$$u = 0, \qquad \operatorname{tr} Y = m.$$

From $tY = \mu I_m$ and $t > 0$, we obtain

$$Y = \frac{\mu}{t} I_m.$$

Taking traces gives

$$m = \operatorname{tr} Y = \frac{\mu}{t} \operatorname{tr}(I_m) = \frac{m\mu}{t},$$

hence $t = \mu$, and therefore $Y = I_m$.

Thus the BKKT system has exactly one solution for each $\mu > 0$, namely

$$x(\mu) = (\mu, 0), \qquad Y(\mu) = I_m.$$

It is smooth in μ. □

Proof of part (B) *of the theorem.* By the preceding lemmas, the family satisfies the prompt's structural assumptions at $x^* = (0, 0)$, has a non-singleton multiplier set, and has analytic center multiplier $Y_a = I_m$. Its central path is explicitly

$$x(\mu) = (\mu, 0).$$

Therefore

$$\dot{x}(\mu) = (1, 0, \ldots, 0)$$

for every $\mu > 0$, so certainly

$$\lim_{\mu \downarrow 0} \dot{x}(\mu) = (1, 0, \ldots, 0).$$

Also

$$\frac{x(\mu) - x^*}{\mu} = (1, 0, \ldots, 0),$$

so the prompt's vector ξ^* equals $(1, 0, \ldots, 0)$. Hence

$$\lim_{\mu \downarrow 0} \dot{x}(\mu) = \xi^*.$$

$\square$

Gap to the full statement

What remains unproved is the full strong reading:

Under the exact weaker Okuno assumptions, without nondegeneracy, does one always have $\dot{x}(\mu) \to \xi^*$?

The prompt gives the secant asymptotic

$$x(\mu) = x^* + \mu\xi^* + o(\mu),$$

but turning this into the tangent asymptotic

$$\dot{x}(\mu) \to \xi^*$$

requires a boundary C^1-estimate for the central path (or, equivalently, an a priori bound for the differentiated symmetric BKKT system near $\mu = 0$). In the nondegenerate regime, Yamashita–Yabe provide exactly such a C^1 continuation. In the weaker regime described in the prompt, that continuation estimate is not supplied in the accessible material I audited, so I do not claim a full proof.

Self red-team

1. Counterexample attempt to the naive secant⇒tangent step. The scalar function

$$q(\mu) = \mu^2 \sin(1/\mu) \qquad (\mu > 0)$$

satisfies $q(\mu)/\mu \to 0$, but

$$q'(\mu) = 2\mu \sin(1/\mu) - \cos(1/\mu)$$

has no limit. So the desired conclusion is *not* a free consequence of the already-known quotient limit.

2. Attempted NSDP counterexample with oscillatory central path. I tried to force a degenerate diagonal NSDP to have

$$x_1(\mu) = \mu, \qquad x_2(\mu) = \mu^2 \sin(1/\mu),$$

by taking $G(x) = x_1 I_2$ and designing f so that $f_{x_2} = 0$ produced the oscillatory branch. This can indeed be arranged with C^2 data if one uses a cubic equation in x_2. However, precisely there the reduced Hessian in the critical x_2-direction vanishes at x^*, so the second-order sufficient condition fails. Thus this attempt does *not* refute the prompt's theorem; it identifies why SOSC matters.

3. Three referee-style objections, and replies.

- *Objection:* In part (A), why does the C^1 branch from Yamashita–Yabe coincide with the given central path?
  Reply: Because both solve the same BKKT system for the same $\mu > 0$, and the prompt assumes uniqueness of the central path.

- *Objection:* In part (B), is the analytic center really I_m?
  Reply: Yes; it is the unique maximizer of $\log\det Y$ under $\operatorname{tr} Y = m$, by AM–GM on the eigenvalues.

- *Objection:* Does the example in part (B) genuinely satisfy the second-order condition uniformly over the whole multiplier set?
  Reply: Yes; the Lagrangian Hessian is independent of Y, and on the critical cone it is exactly $\|v\|^2$.

4. Boundary checks.

- If the central path already extends C^1 to $\mu = 0$, then the claim is immediate; part (A) is consistent with this.

- In the explicit degenerate example, the multiplier set is highly nonunique, so the result is not an artifact of multiplier uniqueness.

- The failed oscillatory construction shows that dropping SOSC can indeed reintroduce secant/tangent mismatch.

Ideas tried

1. **Direct calculus from the quotient limit.**
   Idea: Use $\frac{x(\mu)-x^*}{\mu} \to \xi^*$ to deduce $\dot{x}(\mu) \to \xi^*$.
   Obstruction: False for general C^∞ functions on $(0, \bar{\mu})$; oscillatory counterexamples exist.
   Next step: One must exploit the BKKT structure, not just the quotient limit.

2. **Full proof from the differentiated BKKT system.**
   Idea: Differentiate the symmetric BKKT equations, pass to the limit $\mu \downarrow 0$, and use uniqueness of the Δx-component of the limit system.
   Obstruction: I could not derive, from the prompt alone, the missing a priori estimate that keeps $\dot{x}(\mu)$ under enough control to pass to the limit in full generality.
   Next step: Audit the exact symmetric Newton system in Okuno's paper and prove uniform boundedness (or a weaker compactness statement) for the primal component.

3. **Counterexample construction inside NSDP.**
   Idea: Build a degenerate diagonal NSDP with an oscillatory central-path component.
   Obstruction: Every successful oscillatory construction I found destroyed the second-order sufficient condition at the KKT point.
   Next step: Either prove this obstruction is systematic, or find a more subtle matrix-valued construction where multiplier degeneracy and matrix coupling create oscillations while preserving enhanced SOSC.

Invoked results

Yamashita–Yabe, Lemma 1 (used in part (A)). *Statement.* Under assumptions (A1)–(A4) — namely:

- second derivatives of the data are locally Lipschitz near x^*,
- the second-order sufficient condition holds at x^*,
- strict complementarity holds at x^*,
- the nondegeneracy condition holds at x^*,

there exists a continuously differentiable function

$$\bar{w}(\mu) = (\bar{x}(\mu), \bar{y}(\mu), \bar{Z}(\mu))$$

for $\mu \geq 0$ sufficiently small such that

$$\bar{w}(0) = w^*, \qquad r(\bar{w}(\mu), \mu) = r_S(\bar{w}(\mu), \mu) = 0,$$

and $\bar{X}(\mu) \succ 0$, $\bar{Z}(\mu) \succ 0$ for $\mu > 0$.

Classification. **TOOL**.

Citation. H. Yamashita and H. Yabe, *Local and superlinear convergence of a primal-dual interior point method for nonlinear semidefinite programming*, Mathematical Programming 132(1–2):1–30, 2012, DOI $10.1007/s10107-010-0354-x$. The theorem text was audited on the accessible Optimization Online preprint, and the journal metadata/DOI were audited from the Tokyo University of Science repository entry.

Hypothesis-match audit. In theorem part (A), I explicitly assume exactly these stronger hypotheses. Their variables (x, y, Z), residuals r, r_S, equality map g, and matrix map X are the same objects as the prompt's (x, z, Y), BKKT residuals, h, and G, up to notation. Thus the lemma applies verbatim after renaming symbols.

Self-confidence

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I am confident in the correctness of the two proved affirmative results and in the diagnosis of the missing step for the full weaker statement, but I have not completely resolved the exact strong reading under the full Okuno assumptions.